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International Journal of Psychology
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/pijp20

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Published online: 24 Feb 2007.

To cite this article: Angeliki Gena (2006) The effects of prompting and social reinforcement on establishing social interactions with peers during the inclusion of four children with autism in preschool, International Journal of Psychology, 41:6, 541-554, DOI: 10.1080/00207590500492658

To link to this article: http://dx.doi.org/10.1080/00207590500492658

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INTERNATIONAL JOURNAL OF PSYCHOLOGY, 2006, 41 (6), 541–554

Psychology Press
Taylor & Francis Group

The effects of prompting and social reinforcement on establishing social interactions with peers during the inclusion of four children with autism in preschool

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Inclusion in "regular" schools has become a considerable option for children with autism only in recent years, but we are still far from having a global appreciation of the needs that arise upon the inclusion of children with such a severe disability. The main purpose of the present study was to identify empirically supported procedures that may improve the social interactions of children with autism upon their inclusion in the "regular" preschool. The identification of the needs of preschoolers with autism in Greek schools, as well as the specific skills that would be taught, were based on normative data collected and analysed in a series of prior studies. A long-lasting controversy has been going on between professionals who either advocate for "schools for all children," or stress the importance of making individualized decisions based on each child's needs during the inclusion process. The present study demonstrated that social reinforcement in combination with prompting procedures, provided by a shadow teacher, were effective in increasing the social initiations as well as appropriate responding to peers' initiations of four children with autism during interactions with their classmates in preschool. More importantly, the treatment benefits were obtained in a natural setting, and the initiations and replies were not cliché statements but involved generalized language use appropriate to the social context, and generalized to new therapists.

L'insertion dans les écoles régulières est une option considérable pour les enfants autistiques, mais celle-ci est très récente. Néanmoins, nous sommes encore loin d'avoir une appréciation complète des besoins émergeant de l'insertion des enfants ayant de telles incapacités sévères. Le but principal de la présente étude était d'identifier les procédures soutenues empiriquement qui permettent d'améliorer les interactions sociales des enfants autistiques lors de leur insertion dans les écoles régulières. L'identification des besoins des enfants autistiques d'âge préscolaire dans les écoles grecques, ainsi que les habiletés spécifiques à être enseignées, ont été basées sur des données normatives collectées et analysées par une série d'études antérieures. Une controverse existe depuis longtemps entre les professionnels qui préconisent l'accessibilité à l'école pour tous les enfants et ceux qui plaident en faveur de l'importance de prendre des décisions individualisées basées sur les besoins de chaque enfant dans le processus d'insertion. La présente étude a démontré que le renforcement social en combinaison avec des procédures d'incitation, de la part d'un professeur auxiliaire, étaient efficaces pour l'augmentation des initiations sociales tout comme les réponses appropriées aux initiations des pairs de quatre enfants autistiques durant les interactions avec leurs compagnons de classe au préscolaire. Encore plus important, les bénéfices du traitement ont été obtenus dans des milieux naturels; les initiations et les réponses n'étaient pas des clichés artificiels, mais elles impliquaient l'utilisation généralisée d'un langage adapté au contexte social et généralisé aux nouveaux thérapeutes.

La inclusión en escuelas "regulares" sólo en los últimos años se ha convertido en una opción a considerar para el caso de niños con autismo, pero todavía estamos lejos de lograr una apreciación global de las necesidades que surgen de incluir a niños con limitaciones tan severas. El principal propósito del presente estudio fue identificar procedimientos con respaldo empírico que pudieran mejorar las interacciones sociales de niños con autismo al incluirlos en una escuela preescolar "regular". La identificación de las necesidades de los preescolares con autismo en las escuelas griegas, lo mismo que las habilidades específicas que se les iban a enseñar se basaron en datos normativos recolectados y analizados en una serie de estudios. Ha tenido lugar durante largo tiempo una

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http://www.psypress.com/ijp DOI: 10.1080/00207590500492658

controversia entre los profesionales que están de acuerdo con ''escuelas para todos los niños'' y los que enfatizan la importancia de tomar decisiones individualizadas con base en las necesidades de cada niño durante el proceso de inclusión. El presente trabajo demostró que el reforzamiento social combinado con procedimientos de instigación proporcionados por un maestro auxiliar era efectivo para aumentar la iniciación de interacciones sociales y para responder apropiadamente a las interacciones iniciadas por los compañeritos, en el caso de cuatro niños con autismo estudiados en sus interacciones con compañeros en el preescolar. Aún más importante, vale la pena señalar que los beneficios se obtuvieron en un ambiente natural, las iniciaciones y las respuestas no fueron clichés artificiales, sino que involucraron el uso generalizado de lenguaje apropiado al contexto social y generalizado a nuevos terapeutas.

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INTRODUCTION

Since the 1980s, the issue of inclusion of children with disabilities has provoked a long-lasting controversy among professionals who either advocate for ''schools for all children'' (e.g., Ainscow, 1996; Stainback & Stainback, 1984, 1996; Stainback, Stainback, Courtnage, & Jaben, 1985) or stress the importance of making individualized decisions based on each child's needs during the inclusion process (e.g., Fuchs & Fuchs, 1995; Zigler & Hall, 1995). More recently, however, the types of research questions asked in the area of inclusion attempt ''to determine the differential effects of interventions or settings on children having different characteristics'' (p. 168, Odom, 2002).

The inclusion of children with autism, specifically, has become prevalent in the past few years, because of the increasing number of children diagnosed with autism in the USA (Autism Society of America, 1999; Chakrabarti & Fombonne, 2001; Yeargin-Allsopp et al., 2003), but also because of the complex clinical picture that children with autism present. They often present severe difficulties, not only in the cognitive, but also in the emotional and social domains (e.g., American Psychiatric Association, 1994). Therefore, the importance of social interactions with peers, which has been identified as a critical aspect for child development in general (e.g., Yang, Wolfberg, Wu, & Hwu, 2003), becomes an even more critical issue for children with autism. A regular school setting is often considered the ideal context for the advancement of children's social skills (Mesibov & Shea, 1996). Inclusive educational practices not only realize the right of children with disabilities for a least restrictive placement, but also that the educational and social benefits of inclusion on children with and without disabilities are great, as a growing research base outlines (e.g., McDonnell, 1998). For children with autism, inclusion in regular schools may present a challenge, but it may also play a significant role in their overall adjustment. The breadth and severity of the problems that children with autism present make it unlikely that the majority of those children can benefit from the type of education offered to their typical counterparts. When children with autism receive mainstream educational services that do not address their needs, not only may their cognitive and academic skills fail to advance, but their behaviour management skills may even deteriorate (McClannahan, 1996). In addition, as children with intellectual disabilities, they require specialized services in order to succeed in educational inclusion (Hunt & Goetz, 1997; Laushey & Heflin, 2000; Rogers, 2000). Despite the stand one takes regarding the issue of school inclusion, we would all agree that: (1) the conditions under which it is conducted are critical for a successful outcome and (2) providing effective programming for their educational and social advancement is the most sought-after goal for children with disabilities who are mainstreamed in regular education (e.g., McDonnell, 1998; McDonnell et al., 2003; Robertson, Green, Alper, Schloss, & Kohler, 2003). Thus, upon the inclusion of children with autism in ''regular'' educational settings, we need to address two important questions. First, whether those children would benefit academically and socially in the mainstream and, if not, whether we can provide the proper assistance to the child or make the proper accommodations in the school setting that may contribute to a successful inclusion.

As a large number of studies has demonstrated, the great majority of children with autism lack the skills that would help them benefit from an inclusive educational setting without support (e.g., Evans, Salisbury, Palombaro, Berryman, & Hollowood, 1992; Kamps et al., 2002; Laushey & Heflin, 2000; Ochs, Kremer-Sadlik, Solomon, & Gainer Sirota, 2001). Thus, the research focus in recent years is on identifying intervention procedures that help those children to overcome their social difficulties and begin to interact with peers. Children with autism are challenged in a

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wide array of academic and social skills, but their most prevalent needs appear to be in the area of social integration and in forming relationships with peers during their inclusion in early childhood education settings (Guralnick, 1999; Odom, 2002). Their limited language and social repertoires contribute to their isolation or minimal social interactions with peers and call for support in the area of social engagement (Odom, 2002).

There are numerous intervention studies attempting to identify strategies that may facilitate the inclusion of children with disabilities in "regular" schools, but the great majority of such studies involve children with mild disabilities. On the contrary, there are very few empirical studies that analyse optimal instructional contexts for students with moderate and severe disabilities in general education classrooms (Logan & Malone, 1998). The literature on children with mild disabilities is not necessarily instructive, nor can it be generalized to children with moderate and severe disabilities. One example in which there were clear differences between the progress of children with mild and severe disabilities during the inclusion process was that of the Waldron and McLeskey study (1998), which demonstrated that children with mild disabilities benefited to a greater extent by inclusion in "regular" rather than special education, whereas children with moderate disabilities benefited equally in both types of settings. Therefore, the findings from research involving children with mild disabilities may not necessarily generalize to children with moderate and severe disabilities. What we certainly know, however, is that for students with severe disabilities such as autism, it is imperative, when we consider inclusion in "regular" educational settings, to adopt the necessary teaching practices that address their diverse educational and social needs. Such teaching practices require either direct input from specialists or their input in training teachers or paraprofessionals to use specific interventions efficiently and effectively (e.g., Lovaas, 1987; Schepis, Reid, Ownbey, & Clary, 2003). Thus, for children with autism, the efforts required for successful inclusion are usually extensive and require well-trained professionals (Lovaas, 1987). We will, therefore, focus our attention on studies that address the inclusion of children with autism or other moderate and severe disabilities.

Harrower and Dunlap (2001) classified the types of procedures used to facilitate the inclusion of children with autism into five broad categories. Three of those categories included procedures that enhanced, specifically, their social skills within "regular" education settings. Those categories were classified as: (1) antecedent procedures, (2) peer-mediated interventions, and (3) multicomponent interventions. Peer-mediated procedures appeared to be the most widely used type of intervention and to have the best results.

Peer-mediated interventions, designed to enhance the social skills of children with autism, have been used since the 1980s (e.g., Kamps et al., 1992; Odom, Chandler, Ostrosky, McConnell, & Reaney, 1992; Odom & Strain, 1986), but continue to be used and to be developed to meet the demands of current practices in inclusive education settings (such practices include cooperative learning, peer tutoring, etc.) (e.g., Kamps et al., 2002; Yang et al., 2003; and Hughes, Carter, Hughes, Bradford, & Copeland, 2002). The extensive use of peer-mediated procedures has been attributed to various factors, such as being conducive to the current practices used during inclusion, reducing the financial cost of inclusion as well as the need for continuous one-on-one adult support (Harrower & Dunlap, 2001), at times when the special education support services offered to local national schools are limited (O'Reilly et al., 2002), and promoting maintenance of acquired social skills (e.g., peers may function as mediating stimuli) (Kamps et al., 2002).

Despite the many advantages of peer-mediated procedures, some limitations may also be pointed out with regard to their use toward improving the social skills of children with autism. First, if we take into consideration the importance of early intervention for children with autism (e.g., Fenske, Zalenski, Krantz, & McClannahan, 1985; Lovaas, 1987), the use of peer modelling procedures may not be applicable during the first steps of those children's inclusion in preschools, because for 3- to 4-year-old, typically developing preschoolers, it may be difficult to comprehend and to cooperate in peer tutoring programs, or to use peer-mediating strategies reliably. There is only one study in which such young students were trained as peer tutors, but the tutees were not children with severe disabilities, nor did the peer tutoring goals involve social interactions with verbal exchanges (Robertson et al., 2003). Another issue that needs to be explored with regard to peer-mediated procedures is the quality of interactions that are produced between tutors and tutees as a result of using such procedures. There are a few studies that either examine the relation between teaching procedures and the type of social interactions that they produce (e.g., Cole, Vandercook, & Rynders, 1988) or alert us to the need for qualitative analyses of the types of interactions that develop

between children with and without disabilities (e.g., Evans et al., 1992). In that context, Janney and Snell (1996), suggest that peer-mediated practices should be provided in the context of "cooperation" and "mutual assistance" among all classmates, rather than as special procedures offered exclusively to students with disabilities, which may lead to the differentiation and stigmatization of such students. Finally, a very important outcome of the Kamps et al. (2002) study, which had a very large sample of children with autism, was that, despite the great benefits of peer training on the social interactions among students with autism and their classmates, "no changes were noted for language [of verbal students with autism]."

The second type of intervention strategies aiming to enhance the social skills of children with autism in inclusive settings involve the use of antecedent procedures. They are rather scarce, since the majority of behaviour analytic studies incorporate manipulations of both antecedent and consequential stimuli. Such procedures, when used exclusively, often serve a proactive function, as opposed to targeting directly the modification of specific response classes (Harrower & Dunlap, 2001).

The third type of strategies used to facilitate the inclusion of students with autism incorporate multiple treatment components (Harrower & Dunlap, 2001). There are also very few studies of this type that specifically address social skills training (e.g., Hunt, Farron-Davis, Wrenn, Hirose-Hatae, & Goetz, 1997; Yang et al., 2003). It is worth exploring which of those procedures have been found to be effective. First, a promising approach, not yet extensively researched in the area of social skills, is self-management or self-regulation. Self-management procedures have been very effective in improving the performance of children with developmental disabilities in typical classrooms, but mostly on academic tasks (e.g., Koegel, Harrower, & Koegel, 1999; Wehmeyer, Yeager, Bolding, Agran, & Hughes, 2003). Self-management was used for social skills training in one study, but in conjunction with edible reinforcers (Koegel, Koegel, Hurley, & Frea, 1992). Thus, we may not ascertain to what extent each of the procedures—delivery of primary reinforcers or self-management—attributed to the acquisition of the target responses. In addition, acquired responses did not maintain upon the withdrawal of treatment for two of the subjects.

When using systematic treatment procedures during the inclusion of students with moderate or severe disabilities, the presence of a well-trained shadow teacher, who provided direct support to the child during the inclusion process, was one of the critical variables for enhancing both the academic skills of children with moderate disabilities (e.g., McDonnell et al., 2003) as well as the social skills of children with autism (Schepis et al., 2003).

In summary, from a review of the studies that incorporated multicomponent interventions to enhance the social skills of children with autism in "regular" education settings, we may conclude that there is not sufficient empirical evidence to demonstrate the effectiveness of self-management or self-regulation procedures in improving those skills. Yet, direct support from well-trained staff has been clearly identified as a critical variable for such improvement.

Two other points that were stressed in the literature addressing the inclusion of children with autism are, first, the lack of research in promoting social interactions between children with and without special needs in school playgrounds (Nabors, Willoughby, Leff, & McMenamin, 2001), and second, a lack of studies incorporating experimental methodologies (Schwartz, Sandall, Garfinkle, & Bauer, 1998). The majority of published studies on the inclusion of children with autism used nonexperimental methodologies, such as the paradigm of action research, which, though useful for practitioners, do not allow for the identification of causal relations between treatment manipulations and outcome.

The research questions that arise from the literature discussed in the present review, as well as the desire to explore important variables that may be specific to the Greek culture, led to a series of empirical studies conducted in preschools and kindergartens in Greece. The initial studies of this series have attempted to identify the deficits that children with autism presented upon their inclusion in kindergarten, which prevented them from benefiting from the learning and social opportunities offered in the inclusion settings (Gena & Kymissis, 2001; Gena & Logothetis, 2001; Γενά, 2001, 2002). The most important findings of those studies were the following. (1) Preschoolers of typical development interact in very complex ways and their interactions are quite frequent during free play and recess times. More specifically, content analysis of the interactions of typically developing preschoolers, during play activities, revealed that there were about 25 response categories, with several subcategories of interactions (Gena & Kymissis, 2001). (2) Children with autism lag behind their peers on various areas, including on-task behaviour, responding to

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teachers' requests or questions, etc. Yet, the two areas in which all children invariably demonstrated severe difficulties pertain to the social domain. Making initiations toward classmates, and responding appropriately to their classmates' attempts to interact with them, were the two types of responses that children with autism either did not have at all in their repertoires, or demonstrated minimally. (3) A girl with autism, who was mainstreamed in kindergarten, made excellent progress in all the areas mentioned above and particularly in the social domain. (4) The variables that may have contributed to that progress include intensive behaviour analytic intervention at home and close supervision and guidance in the inclusion setting provided by a shadow teacher with a behaviour analytic orientation (Gena & Logothetis, 2001). The above findings have stimulated the interest to explore the effects of the intervention in an experimental manner. Thus, the purpose of the present study was to assess whether, for children with autism who receive intensive behaviour analytic intervention at home, a multicomponent intervention would be effective in helping them to improve the types of social interactions with peers that require generalized language use. Specifically, the purposes of the present study were formulated as follows: (1) to analyse whether the combination of antecedent strategies (i.e., prompting), social reinforcement, and time out from reinforcement contingent on noncompliant behaviour would increase the frequency of social responding that involves generalized language use (i.e., making social initiations and responding to initiations made by peers) in four preschoolers with autism; (2) to provide an experimental analysis of the effectiveness of treatment using a multiple baseline across subjects experimental design; (3) to investigate whether the intervention would be successful with young children, around the age of 4 years; (4) to promote response generalization of the target responses by using diverse and loose training in the sense of not teaching cliché initiation or reply statements, but rather using types of contextually appropriate responses when prompting from the shadow teacher was required; (5) to foster the development of social skills in a natural setting by selecting the preschool playground as the training context, which was considered important since children with autism are often excluded from interactions with peers in settings, such as the playground, where most children their age typically engage in high rates of social exchanges (Nabors et al., 2001); (6) to use a method of direct observation, which may be viewed as a performance-based assessment, which is considered a more accurate type of assessment to evaluate treatment outcome (Gettinger & Callan Stoiber, 1998); and (7) to assess fluency of appropriate affective responding by comparing the level of social responding of children with autism to that of their typical classmates in the target areas.

METHOD

Participants

The participants of this study were four children with autism. We will also refer to normative data for comparison purposes, without providing a detailed description of the typically developing peers, since it has already been provided in a previous publication (Gena & Kymissis, 2001). There were 12 typically developing preschoolers—3 per child with autism—aged from 3 years and 10 months to 4 years and 7 months. They were matched for gender and age to the children with autism and were selected randomly from the preschools that the children with autism attended.

The target children were two girls and two boys with the diagnosis of autistic disorder. Their diagnosis was based on public hospital evaluations, as well as evaluations of the centre where they received intervention (the Athenian Centre for Child Development and Education), both of which were located in Athens, Greece. Other than having the diagnosis of autism, all four children had been excluded from several private and public preschools because of their behavioural and learning difficulties, according to their parents' reports.

All four participants received behaviour analytic intervention, mostly one-on-one, in dyads or small groups prior to and during their inclusion in "regular" preschool. At the beginning of the study, Lena was 4 years old and had been in therapy for 7 months. Her therapeutic programme included, on the average, 7 hours per week one-on-one, in-home, and intervention by experienced therapists. In addition, her mother received training and was very competent in carrying out Lena's therapeutic goals. Lena's attendance started at a private preschool with 1 hour per day and gradually increased to a full-day attendance (4 hours).

Eva was 4 years and 5 months old at the beginning of the study, and had been in therapy for 3 months receiving one-on-one, in-home intervention for an average of 9 hours weekly. Eva attended a public preschool and her attendance schedule was similar to Lena's.

Andrew was 4 years and 2 months old and had been in therapy for 4 months prior to inclusion. Initially, he attended a day programme for children with autism, where he received behaviour-analytic, mostly one-on-one intervention for 24 hours per week, which progressively decreased to approximately 10–12 hours weekly. His inclusion took place in a private preschool starting with 2 hours per week and increased to 4 hours per day.

At the beginning of the study Cory was 4 years old and had been in therapy for 2 months prior to inclusion. His inclusion took place in a public preschool starting with 4 hours per day with the shadow teacher, whose presence was faded, progressively, to 6 hours per week.

Regarding the participants' overall progress in therapy, it is worth mentioning that all four started out with limited language (mostly one-word utterances), severe delays in communication, inappropriate affective behaviour, and aberrant behaviour. Yet, shortly after intervention, they all showed great improvement in their use of expressive and communicative language, in motivation and in compliance toward parents and therapists. Additionally, behavioural problems, such as temper tantrums and stereotypic behaviour, had decreased dramatically for all the participants prior to their inclusion in preschool. Even though the participants were considered severely disabled and nontestable by standardized assessments before treatment, their intellectual functioning reached age-appropriate levels in the course of treatment, but they continued to demonstrate severe social and emotional difficulties, which led to the need for additional help during their attendance in "regular" preschool. No standardized measures of their intellectual, emotional, or social abilities can be provided, since there are no psychometric tests available for the Greek population of their age group.

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Settings and therapists

All data were collected in the inclusion preschool settings, which were located in three middle-class areas of Athens, in Greece. Data were collected during semistructured or free-play activities, because they were considered ideal for social interactions between children, unlike structured activities, which typically require attending and responding mainly to the teacher.

During both baseline and treatment, well-trained shadow teachers were assigned to the participants (one teacher per child). The shadow teachers were experienced therapists who had worked for several years with children with autism, but received additional training for a minimum of 3 months to familiarize themselves with the intervention implemented for the purposes of the present study. Each one of the shadow teachers was familiar with the children she supervised in the inclusion setting. All shadow teachers were college graduates who had majored either in psychology or in education.

Inter-observer agreement data were collected by the shadow teachers, three other therapists of children with autism, who were college students, and the author of the present study. The college students received training on the data collection procedures of the present study.

Definitions of the target responses and collection of normative data

Student initiations toward classmates were defined as statements and gestures that the participants made toward classmates. Those statements and gestures had to be socially appropriate, relevant to the context of the interaction, independent of prompting by the shadow teacher, and made in the absence of inappropriate or stereotypic behaviour.

Student reply statements were defined as verbal and nonverbal responses that followed initiations made by classmates, were relevant to those initiations, were not prompted by the shadow teacher, and were not accompanied by inappropriate or stereotypic behaviour. Initiations and reply statements that were prompted by the preschool teacher or by classmates were scored as independent.

To determine the types of initiation statements that we would teach to children with autism, we used data of typically developing preschoolers. Normative data were collected twice, at the beginning and at the end of the school year, with approximately 8 to 9 months elapsing between the two observation periods. There were a total of 10 half-hour observation periods per child, 5 in the beginning and 5 at the end of the school year. The conditions under which normative data were collected were similar to the conditions under which baseline and treatment data were collected. Namely, independent observers recorded verbatim the initiation statements, as well as the reply statements toward classmates that the 12 typical preschoolers made during free-play or semistructured activities (see Gena & Kymissis, 2001, for further details). Specifically, we analysed the initiation statements of 12 preschoolers matched for gender and age to the target preschoolers.

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Independent observers recorded initiations and reply statements addressing classmates during free-play or semistructured activities. The analysis of those data yielded over 20 categories and several subcategories of initiations, but only the 5 most frequently used types of initiations were selected for the purpose of training children with autism. We considered that teaching more than five types could be overwhelming for children who do not initiate interactions at all. The five categories of initiations were as follows: (1) asking questions either to obtain information or to provoke competition, (2) demonstrating affective responses expressed verbally or with gestures, (3) asking a classmate to perform an action or to make a statement, (4) announcing information about their own and others' actions at the present or at a remote time, and (5) inviting classmates to participate in a common activity. Definitions of the response categories selected for the purposes of the present study and examples from each category are presented in Table 1.

Experimental design, data collection, conditions, and procedure

A multiple baseline across subjects single-case experimental design was used to assess the efficacy of the treatment procedure. During baseline, the shadow teachers were present and guided the participants to attend and participate in activities, but neither prompted nor reinforced target social responses.

During treatment, the shadow teachers used prompting and social reinforcement contingent upon target responses, as well as instruction following. Prompting was defined as any kind of verbal prompt or manual guidance provided by

TABLE 1
Response categories of initiations toward peers

Response categories	*Definitions and examples*
1. Asking questions	Posing questions that pertain to the following areas: (a) seeking information; (b) provoking competition. E.g., "Do you know what we have to do with this worksheet?" or "Can you write 'A' like me?"
2. Displaying affective behaviour	Showing affection toward a peer either verbally or with gestures: (a) with gestures; (b) verbally; (c) offering something with affection (e.g., "Here, you can have my painting"); (d) greeting with affection (e.g., "Hi, I missed you"); (e) asking for one's friendship (e.g., "I want you to be my best friend"); (f) seeking reassurance for one's friendship (e.g., "Am I still your best friend?"); (g) showing enthusiasm (e.g., "I love to sing along with the teacher"); (h) showing sympathy (e.g., "Feel better, Stephania"); (i) showing anger (e.g., "Don't you dare take my marker again!").
3. Giving commands	Making statements that require classmates to perform an action or to make a statement. E.g., "Put this tissue in the garbage" "Tell her what your favorite colour is."
4. Announcing information	Providing information regarding: (a) own actions at present time; (b) own actions at remote time (past or future); (c) others' actions. E.g., "Look, I drew an airplane" or "I am going to go to the playground this afternoon."
5. Inviting classmates	Inviting classmates to participate in activities other than play activities. E.g., "Sit next to me" or "Will you come to my birthday party?"
6. Other	Making any type of initiation that was not included in the five categories on which data were specifically collected. E.g., making welcoming or goodbye statements upon the entrance of a classmate or upon his own entrance or departure from the preschool, or giving a compliment to a classmate.

the shadow teacher for the child with autism to engage in either one of the target responses. For example, to encourage social initiations toward classmates, the shadow teacher might have asked the student to say something like: ''Let's play in the backyard,'' or ''Look at my doll, isn't she pretty?''

Social reinforcement was provided contingent upon the participants' spontaneous social initiations, as well as upon their independent responses to peer initiations. Social reinforcement included verbal statements, such as ''you are doing a great job!'' a pat on the back, and other forms of social approval.

''Instruction following'' was used as a behaviour management procedure to reduce the frequency of aberrant behaviour such as noncompliance, inappropriate behaviour toward classmates or the teacher, and having temper tantrums. During ''instruction following'' the participants were removed from the classroom or the playground and were accompanied to an empty classroom or the preschool kitchen, where they had to follow simple directions, such as to complete a puzzle, to answer social questions given by the shadow teacher, etc. In addition, the shadow teacher reminded the child of their agreement by saying, ''When you are ready, we will go back to your friends.'' The exit criterion from the instruction following procedure was that the child would follow three consecutive instructions without crying or fussing.

Data on initiation statements were collected specifically for each one of the five aforementioned categories. The data collection sheet for initiations included boxes corresponding to each one of the response categories, as well as a separate box for the category of ''other.'' Tally marks under each category were used to indicate the occurrence of initiations per response category. The sum of those tally marks per session was plotted as one data point.

Data on reply statements were collected on a separate data sheet, on which initiation statements of classmates were reported followed by a ''+'' or ''–'', indicating the presence or absence, respectively, of an appropriate reply statement provided by the target student. The percentage of appropriate reply statements was calculated by dividing the number of appropriate statements by the total number of appropriate and inappropriate statements and multiplying the quotient by 100. The percentage per session was plotted as one data point.

Generalization data were collected only for two of the participants, because school regulations of the other two preschools did not allow for the absence of the shadow teacher. During generalization sessions, the target child was observed in the absence of his shadow teacher and no type of intervention was provided. The novel therapist was present and collected data discretely, but did not interact with the child.

Baseline, intervention, and generalization data were collected over several months: 9 months for Eva, 13 months for Lena, and 15 months for Andrew and Cory. The intervention was ongoing, but the total observation time for data collection purposes ranged from 12 to 22 hours per child across experimental conditions.

Fidelity of treatment measures were collected on at least 70% of sessions per experimental phase and per child and indicated that research staff implemented experimental procedures 100% as intended across all assessed sessions. The high accuracy of implementing treatment procedures may be attributed to the fact that all the shadow teachers were very experienced in using the experimental procedures, since they had used them, for clinical purposes, for at least 1 year prior to the present research study.

Inter-observer agreement

Inter-observer agreement data were collected on at least 33% of observations for each child and during each experimental condition. The frequency ratio agreement method (Kazdin, 1982) was used to assess agreement between observers for replies and initiations. Specifically, the smaller total number of initiations scored by one observer was divided by the larger number of initiations scored by the second observer and the quotient was multiplied by 100. Respectively, for reply statements, the smaller percentage of appropriate reply statements was divided by the larger percentage of inappropriate reply statements and the quotient was multiplied by 100. Inter-observer agreement across all target responses ranged from 85–100%, with an average of 95% agreement. The same procedure was used to assess the dependent measures and agreement was invariably at 100%.

Social validation

All four participants refused to attend preschool prior to intervention and had severe temper tantrums to avoid going to school. They all preferred to engage in one-on-one, in-home therapeutic sessions, rather than having to go to preschool. Shortly after the intervention, however, attending preschool was

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Number of Initiations Toward Classmates

Therapist: _____

Date _____ Student: _____

Time	Initiation Categories					
	Questions	Affective Re.	Commands	Announcements	Invitations	Other

Figure 1. Data sheet for initiations toward classmates.

preferred to one-on-one therapeutic sessions, and going to preschool became highly valued by the participants (e.g., they wanted to go to preschool even when sick). In addition, both parents and preschool teachers reported that the intervention procedure was both highly necessary and critical for the child's social adjustment. Finally, the target responses may be considered socially valid, because their selection was based on normative data.

RESULTS

A detailed description of the normative data pertaining to initiations and reply statements made

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by preschoolers of typical development has already been published (see Gena & Kymissis, 2001). Therefore, we will only provide a brief account of those data as a means of comparing them to the data collected for the three preschoolers with autism. During the first observation period, the number of social initiations made by classmates of typical development toward peers ranged from 1 to 52 with an average of 22 per half-hour observation, as compared to 14 to 79 with an average of 36 when they were measured 9 months later. The percentage of reply statements during the first observation period ranged from 45–100% with an average of 78%, whereas the corresponding percentages 9 months later ranged from 70–92% with an average of 84% per half-hour observations.

The number of initiations that children with autism made toward their classmates is presented in Figure 2. Lena, during baseline, made 0–4 initiations with an average of 3 initiations per half-hour observation period. During intervention, however, her initiations increased dramatically with a range of 9–55, showing no overlap with baseline, and an average of 26 per half hour. Eva, Andrew, and Cory did not initiate any interactions with peers during baseline, but during intervention their initiations also showed a dramatic increase. Eva's initiations ranged from 3–31, again with no overlap with baseline points, with an average of 14 initiations during half-hour observation periods. Andrew's initiations ranged from 2–53, with an average of 20 per half-hour session, and Cory's ranged from 4–23 with an average of 14 initiations.

Figure 3 depicts the percentages of replies that children with autism provided to their classmates' initiations. Lena's replies to her classmates' initiations ranged from 0–25% with an average of 15% during baseline, as compared to a range of 42–93% with an average of 72% during treatment. Eva did not reply to any initiations made by her classmates during baseline; neither did Andrew or Cory. Following intervention, however, their replies increased greatly. Eva's replies ranged from 25–71% with an average of 54% during half-hour observation periods, Andrew's from 50–90% with

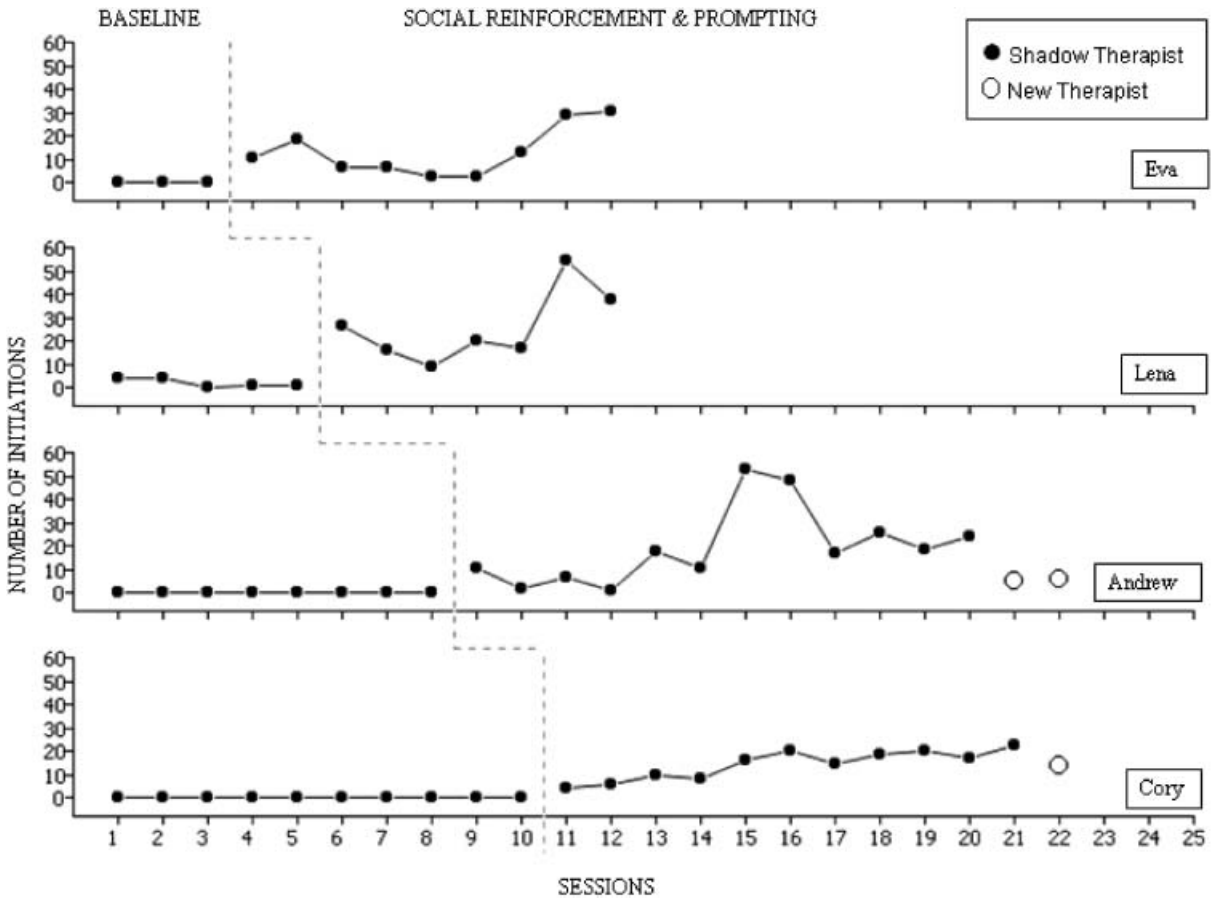


Figure 2. Number of initiations of four children with autism during their inclusion in preschool. The closed circles depict initiations in the presence of a shadow teacher and the open circles depict initiations in the presence of a new therapist.

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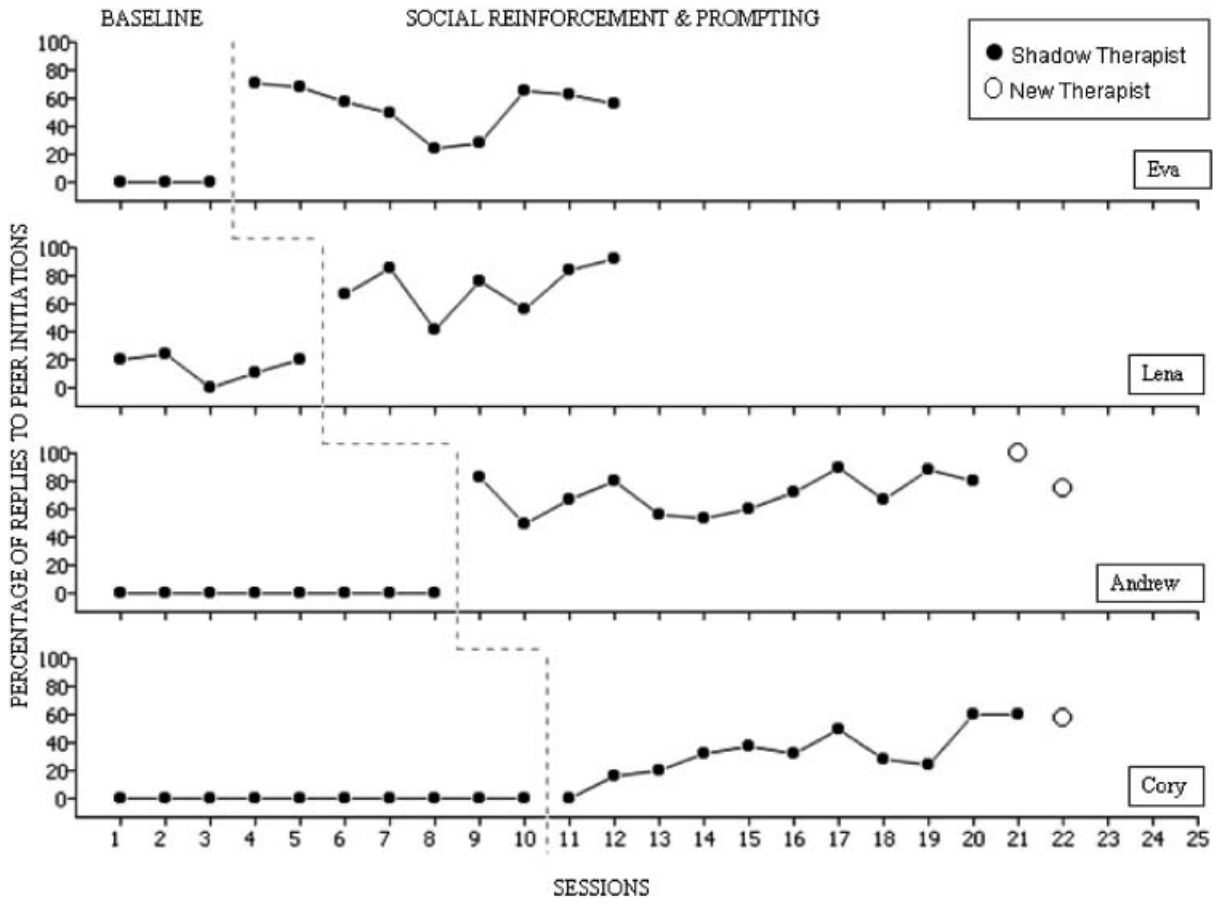


Figure 3. Percentage of reply statements to peer initiations of four children with autism during their inclusion in preschool. The closed circles depict replies in the presence of a shadow teacher and the open circles depict replies in the presence of a new therapist.

an average of 71%, and Cory's from 0–60% with an average of 33%. Overall, it is worth noting that there was no overlap of baseline and treatment points for any of the participants or the categories of target responses.

With regard to generalization across people, Andrew made 5 initiations during the first session and 6 during the second, versus 0 in baseline, but certainly fewer than the initiations he made in the presence of his shadow teacher. Nevertheless, his replies to peer initiations ranged from 75–100%, which are certainly higher than the 0% in baseline, but also as high as the percentage of replies during treatment. Cory's generalization data (14 initiations and 58% replies) are higher than his baseline data (0 initiations and 0% replies) and at approximately the same level as his performance during intervention.

"Instruction following," the behaviour reduction procedure used to reduce temper tantrums and noncompliance, was used as many times as the participants were noncompliant or had temper tantrums. It was implemented, during baseline, 7 times for Lena, 4 for Eva, 2 for Andrew, and 6 times for Cory. There was no need to implement the procedure during treatment.

DISCUSSION

The results of the present study are in agreement with prior research indicating that, during school inclusion, children with autism demonstrate limited communicative or social interactions with their classmates. Specifically, during baseline, three of the participants neither took initiative to interact with their classmates, nor did they respond appropriately when addressed by their classmates. Only one of the participants demonstrated some appropriate interactions, but minimally. Despite their social aloofness, however, they showed dramatic improvement in the target areas. This leads us to conclude that children with autism who receive systematic behaviour analytic intervention from early on and have reached age-appropriate intellectual functioning, like the

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participants of the present study, may overcome major social difficulties and consequently establish appropriate social interactions with their classmates when provided with systematic treatment by well-trained shadow teachers. Specifically, an intervention including social praise contingent upon the target behaviour and prompting as a correction procedure may be effective in increasing the social initiations toward classmates as well as responding to peer initiations. In addition, compliance to the shadow teacher's instruction and abstaining from inappropriate behaviour, such as temper tantrums, may be critical in obtaining the treatment outcome of the present study. The "instruction following" procedure selected for the purposes of the present study was only used a few times and was sufficient for achieving compliance and for extinguishing temper tantrums.

The increase in appropriate social responding that the participants of the present study noted may not be attributed to maturational factors for several reasons: (1) none of the participants demonstrated improvement in the target responses before intervention was introduced; (2) there was an increase in target responding precisely at the point in time when treatment was introduced for each participant; (3) even though the participants had attended preschool for several months, prior to receiving intervention, they had made no progress in the social domain and had to be excluded from school due to aberrant behaviour; and (4) their typically developing peers did not demonstrate such dramatic progress in the target areas during the school year.

The effectiveness of treatment in the present study may be attributed to several factors. First, the four children with autism who participated in this study, during their inclusion in preschool, were not simply placed in physical proximity to their typically developing counterparts, but were provided with systematic treatment used by competent therapists, well-trained in the use of behaviour analytic procedures. In addition, the intervention in preschool was not the only treatment that the participants received. Before, as well as during, their inclusion, they also received extensive treatment addressing their needs across all areas of development. Furthermore, control of their inappropriate behaviour was attained with minimal intervention, most likely because the shadow teachers were also serving as home therapists for the children and had established a good working relationship with them prior to their inclusion in preschool. Finally, because the participants were trained to use socially validated initiations, it is very likely that their social responses were also reinforced by natural contingencies (e.g., peer attention and play with peers), rather than by the social reinforcement provided by shadow teacher alone.

The clinical significance of the present study may be high, but there are several limitations and there is certainly a need for further experimental investigation in this area. First, although the internal validity, as well as the reliability, of the results is not in doubt, since a direct measure of the dependent variable was employed (data were collected through direct observations) and the agreement between observers was high, the small sample size and the nonrandom selection of the sample limit the external validity of the findings. Thus, the findings generalize specifically to the sample of children with autism who reach normal intellectual functioning and receive extensive behaviour analytic intervention, rather than the entire population of children with autism. Another limitation of the present study is that stimulus generalization was assessed in only one parameter (across new people) and for only two of the four participants, because of practical considerations. It would be interesting for future research to test for maintenance of the treatment outcome, in the absence of a shadow teacher, at different points in time during intervention, in order to identify a more parsimonious treatment package. For example, it may be sufficient for shadow teachers to limit their presence to progressively fewer hours, rather than to accompany the student to preschool throughout the day for several months. Other types of manipulations could be identified that would lead to a less intrusive intervention or fewer hours of intervention.

Despite the limitations of the present study, we may conclude that the intervention outcome was socially validated by both the parents and the teachers of the participants, who characterized the intervention as both important and necessary for the social adjustment of children with autism in "regular" preschool. In addition, the intervention took place in a natural setting (a "regular" preschool) and the participants interacted with their classmates in a generalized manner and were trained loosely. That is, they did not learn cliché initiations or to respond to specific initiations from their classmates. Rather, they learned to make initiations and to respond to peer initiations in a socially acceptable manner and in various social contexts (e.g., during group games, group activities, while playing with a variety of toys, etc.). It is also important that the preschool teachers were not burdened with making special accommodations

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for the children with autism, since it was the participants who learned to adjust to preschool routines. For cultures in which the inclusion of children with autism is a relatively new and uncommon practice, making the inclusion process as unobtrusive as possible is important, because it is likely to help teachers and other school personnel to have a more accepting attitude toward inclusion.

The present study is only a small step toward dealing with the complex issues arising from the inclusion of children with disabilities. It sheds some light on the diversity of social skills that children with autism need to acquire in order to be "fluent" in interacting with their classmates. Yet, we are far from identifying the spectrum of criteria that a child and a school have to meet in order for the inclusion of children with autism to be successful and meaningful. The issues of inclusion concern not only children with disabilities, but also the entire school community.

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